$a = -4.305\ \text{m s}^{-2} = 4.3\ \text{m s}^{-2}$ (downwards)

25 **C** When wire Y is perpendicular to wire X (current into the plane of the paper),

$B_Y = \dfrac{\mu_o I}{2\pi d} = B$ (points upwards)

$B_X = \dfrac{\mu_o I}{2\pi d} = B$ (points out of plane)

$B_{\text{resultant}} = \sqrt{B^2 + B^2} = \sqrt{2}\,B$